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import CoolProp.CoolProp as CP
import ipywidgets as widgets
from IPython.display import display, clear_output
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# --- 1. Configuration & Constants ---
fluids = ['Water', 'R134a', 'Air', 'Nitrogen', 'CO2', 'Ammonia',
          ↪ 'Ethanol', 'Methane']

# Input pairs (CoolProp Keys)
input_pairs = {
    'Temperature (K) & Pressure (Pa)': 'TP',
    'Temperature (K) & Quality (0-1)': 'TQ',
    'Pressure (Pa) & Quality (0-1)': 'PQ',
    'Pressure (Pa) & Enthalpy (J/kg)': 'PH',
    'Pressure (Pa) & Entropy (J/kg/K)': 'PS',
    'Temperature (K) & Density (kg/m3)': 'TD',
}

# --- 2. Helper Functions ---

def convert_units(data_dict, unit_system):
    """Converts a dictionary of SI properties to the target unit
    ↪ system."""
    d = data_dict.copy()

    if unit_system == 'SI':
        # Convert J to kJ for readability
        d['P'] = d['P [Pa]'] / 1000.0 # kPa
        d['T'] = d['T [K]']
        d['v'] = d['v [m3/kg]']
        d['u'] = d['u [J/kg]'] / 1000.0 # kJ/kg
        d['h'] = d['h [J/kg]'] / 1000.0 # kJ/kg
        d['s'] = d['s [J/kg/K]'] / 1000.0 # kJ/kg-K
        return d, {'P': 'kPa', 'T': 'K', 'v': 'm3/kg', 'u':
          ↪ 'kJ/kg', 'h': 'kJ/kg', 's': 'kJ/kg-K'}

    elif unit_system == 'Imperial':
        # Conversions
        d['P'] = d['P [Pa]'] * 0.000145038 # psia
        d['T'] = (d['T [K]'] - 273.15) * 1.8 + 32 # F
        d['v'] = d['v [m3/kg]'] * 16.0185 # ft3/lb
        d['u'] = d['u [J/kg]'] * 0.000429923 # Btu/lb
        d['h'] = d['h [J/kg]'] * 0.000429923 # Btu/lb
        d['s'] = d['s [J/kg/K]'] * 0.000238846 # Btu/lb-R
        return d, {'P': 'psia', 'T': 'F', 'v': 'ft3/lb', 'u':
          ↪ 'Btu/lb', 'h': 'Btu/lb', 's': 'Btu/lb-R'}

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def get_phase_desc(fluid, T, P):
    try:
        T_crit = CP.PropsSI('Tcrit', fluid)
        P_crit = CP.PropsSI('Pcrit', fluid)

        if T > T_crit and P > P_crit:
            return "Supercritical"

        try:
            Psat = CP.PropsSI('P', 'T', T, 'Q', 0, fluid)
            if abs(P - Psat) < 1e-1:
                return "Saturation"
            elif P > Psat:
                return "Compressed Liquid"
            else:
                return "Superheated Vapor"
        except:
            return "Fluid"
    except:
        return "Unknown"

def calculate_state(fluid, pair_key, val1, val2):
    inputs = {'TP': ('T', 'P'), 'TQ': ('T', 'Q'), 'PQ': ('P',
    ↪ 'Q'),
              'PH': ('P', 'H'), 'PS': ('P', 'S'), 'TD': ('T',
    ↪ 'D')})

    param1_name, param2_name = inputs[pair_key]

    try:
        # Calculate in SI Base Units
        T = CP.PropsSI('T', param1_name, val1, param2_name,
    ↪ val2, fluid)
        P = CP.PropsSI('P', param1_name, val1, param2_name,
    ↪ val2, fluid)
        D = CP.PropsSI('D', param1_name, val1, param2_name,
    ↪ val2, fluid)
        H = CP.PropsSI('H', param1_name, val1, param2_name,
    ↪ val2, fluid)
        S = CP.PropsSI('S', param1_name, val1, param2_name,
    ↪ val2, fluid)
        U = CP.PropsSI('U', param1_name, val1, param2_name,
    ↪ val2, fluid)
        Q = CP.PropsSI('Q', param1_name, val1, param2_name,
    ↪ val2, fluid)
        v = 1.0 / D if D != 0 else 0
    
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    if 0 <= Q <= 1:
        phase = f"Two-Phase (x={Q:.4f})"
    else:
        phase = get_phase_desc(fluid, T, P)
        Q = np.nan

    return {
        'T [K]': T, 'P [Pa]': P, 'v [m3/kg]': v, 'u [J/kg]':
        ↪ U,
        'h [J/kg]': H, 's [J/kg/K]': S, 'Quality (x)': Q,
        ↪ 'Phase': phase
    }
except Exception as e:
    return {'Error': str(e)}

# --- 3. Visualization Functions ---

def plot_diagrams(fluid, state1, state2, unit_system):
    """Plots both T-s and P-v diagrams side-by-side."""

    # Setup Data for Dome
    T_crit = CP.PropsSI('Tcrit', fluid)
    T_min = CP.PropsSI('Tmin', fluid)
    T_range = np.linspace(T_min, T_crit - 0.5, 50)

    # Calculate Saturation Lines (SI)
    s_liq_si = [CP.PropsSI('S', 'T', t, 'Q', 0, fluid) for t in
    ↪ T_range]
    s_vap_si = [CP.PropsSI('S', 'T', t, 'Q', 1, fluid) for t in
    ↪ T_range]
    v_liq_si = [1/CP.PropsSI('D', 'T', t, 'Q', 0, fluid) for t
    ↪ in T_range]
    v_vap_si = [1/CP.PropsSI('D', 'T', t, 'Q', 1, fluid) for t
    ↪ in T_range]
    p_sat_si = [CP.PropsSI('P', 'T', t, 'Q', 0, fluid) for t in
    ↪ T_range]

    # Convert Dome Data if needed
    if unit_system == 'Imperial':
        # T-s conversions
        T_plot = [(t - 273.15)*1.8 + 32 for t in T_range]
        s_liq = [s * 0.000238846 for s in s_liq_si]
        s_vap = [s * 0.000238846 for s in s_vap_si]
        # P-v conversions
        p_plot = [p * 0.000145038 for p in p_sat_si]
        v_liq = [v * 16.0185 for v in v_liq_si]
        v_vap = [v * 16.0185 for v in v_vap_si]

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        units = {'T': 'F', 'P': 'psia', 'v': 'ft3/lb', 's':
↪ 'Btu/lb-R'}
    else:
        T_plot = T_range
        s_liq = [s/1000 for s in s_liq_si]
        s_vap = [s/1000 for s in s_vap_si]
        p_plot = [p/1000 for p in p_sat_si] # kPa
        v_liq = v_liq_si
        v_vap = v_vap_si
        units = {'T': 'K', 'P': 'kPa', 'v': 'm3/kg', 's':
↪ 'kJ/kg-K'}

    # Prepare Figure
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

    # --- Plot 1: T-s Diagram ---
    ax1.plot(s_liq, T_plot, 'k-', alpha=0.5, label='Sat. Liq')
    ax1.plot(s_vap, T_plot, 'k--', alpha=0.5, label='Sat. Vap')
    ax1.set_xlabel(f"Entropy ({units['s']})")
    ax1.set_ylabel(f"Temperature ({units['T']})")
    ax1.set_title(f"T-s Diagram ({fluid})")

    # --- Plot 2: P-v Diagram ---
    ax2.plot(v_liq, p_plot, 'k-', alpha=0.5)
    ax2.plot(v_vap, p_plot, 'k--', alpha=0.5)
    ax2.set_xlabel(f"Specific Volume ({units['v']})")
    ax2.set_ylabel(f"Pressure ({units['P']})")
    ax2.set_xscale('log') # Log scale for volume is essential
    ax2.set_title(f"P-v Diagram ({fluid})")

    # Plot Points
    for state, color, label in [(state1, 'r', '1'), (state2,
↪ 'b', '2')]:
        if 'Error' not in state:
            # Convert single state point
            s_conv, _ = convert_units(state, unit_system)

            # T-s Point
            ax1.plot(s_conv['s'], s_conv['T'], marker='o',
↪ color=color, label=f"State {label}")

            # P-v Point
            ax2.plot(s_conv['v'], s_conv['P'], marker='o',
↪ color=color, label=f"State {label}")

    ax1.legend()
    ax1.grid(True, alpha=0.3)
    ax2.grid(True, alpha=0.3, which="both")

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plt.tight_layout()
plt.show()

# --- 4. Interface & Logic ---

w_fluid = widgets.DropDown(options=fluids, value='Water',
    ↪ description='Fluid:')
w_units = widgets.ToggleButtons(options=['SI', 'Imperial'],
    ↪ value='SI', description='Output Units:')

# State 1 Widgets
w_pair1 = widgets.DropDown(options=input_pairs, value='TP',
    ↪ description='State 1 Inputs:')
w_val1_1 = widgets.FloatText(description='Prop 1:', value=300)
w_val1_2 = widgets.FloatText(description='Prop 2:',
    ↪ value=101325)

# State 2 Widgets
w_pair2 = widgets.DropDown(options=input_pairs, value='TP',
    ↪ description='State 2 Inputs:')
w_val2_1 = widgets.FloatText(description='Prop 1:', value=400)
w_val2_2 = widgets.FloatText(description='Prop 2:',
    ↪ value=101325)

btn_calc = widgets.Button(description='Calculate',
    ↪ button_style='primary')
out_display = widgets.Output()

def on_calc_clicked(b):
    with out_display:
        clear_output()

        # Calculate Base States (SI)
        s1 = calculate_state(w_fluid.value, w_pair1.value,
    ↪ w_val1_1.value, w_val1_2.value)
        s2 = calculate_state(w_fluid.value, w_pair2.value,
    ↪ w_val2_1.value, w_val2_2.value)

        if 'Error' in s1: print(f"Error State 1:
    ↪ {s1['Error']}"); return
        if 'Error' in s2: print(f"Error State 2:
    ↪ {s2['Error']}"); return

        # Convert for Display
        sys = w_units.value
        s1_disp, u_labels = convert_units(s1, sys)
        s2_disp, _ = convert_units(s2, sys)

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# Create Table Data
props = ['T', 'P', 'v', 'u', 'h', 's']
prop_names = [f"Temperature [{u_labels['T']}]",
↪ f"Pressure [{u_labels['P']}]",
                f"Spec. Vol [{u_labels['v']}]", f"Int.
                ↪ Energy [{u_labels['u']}]",
                f"Enthalpy [{u_labels['h']}]", f"Entropy
                ↪ [{u_labels['s']}]"

data = {'Property': prop_names + ['Quality (x)',
↪ 'Phase']}
data['State 1'] = [s1_disp[p] for p in props] +
↪ [s1['Quality (x)', s1['Phase']]
data['State 2'] = [s2_disp[p] for p in props] +
↪ [s2['Quality (x)', s2['Phase']]

df = pd.DataFrame(data)

# Calculate Delta
deltas = []
for i in range(len(props)):
    deltas.append(data['State 2'][i] - data['State
↪ 1'][i])
deltas.extend(['', '']) # No delta for Q or Phase
df['Delta (2-1)'] = deltas

# Display Table
display(df.style.format(precision=4, subset=['State 1',
↪ 'State 2', 'Delta (2-1)']))

# Plot Diagrams
plot_diagrams(w_fluid.value, s1, s2, sys)

# Reference State Note
display(widgets.HTML(f"""
<div style="background-color: #e7f3fe; border-left: 6px
↪ solid #2196F3; padding: 10px; margin-top: 10px;">
    <strong>Important Note on Reference States:</strong><br>
    Values for Energy (u, h) and Entropy (s) are calculated
↪ relative to a reference state where h=0 and s=0.
    CoolProp often uses the IIR standard (saturated liquid
↪ at 0°C) or NBP (Normal Boiling Point) depending on the
↪ fluid.
    <br><strong>Warning:</strong> The <em>absolute</em>
↪ values shown here may differ from your textbook tables if
↪ the reference states do not match.

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        However, the <strong>Difference (Delta)</strong> values
↪ should match closely, as differences are independent of the
↪ reference state.
        </div>
        """)

btn_calc.on_click(on_calc_clicked)

# Layout
ui = widgets.VBox([
    widgets.HTML("<h2>Thermodynamic Property Calculator</h2>"),
    widgets.HBox([w_fluid, w_units]),
    widgets.HTML("<i><b>Note:</b> Please enter input values in
↪ SI Units (K, Pa, J/kg). Output will convert based on
↪ toggle.</i>"),
    widgets.HTML("<hr>"),
    widgets.HBox([
        widgets.VBox([widgets.HTML("<b>State 1</b>"), w_pair1,
↪ w_val1_1, w_val1_2]),
        widgets.VBox([widgets.HTML("<b>State 2</b>"), w_pair2,
↪ w_val2_1, w_val2_2])
    ]),
    widgets.HTML("<br>"),
    btn_calc,
    out_display
])

display(ui)

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VBox(children=(HTML(value='<h2>Thermodynamic Property Calculator</h2>'), HTML(value='<div>

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